

Quentin Guignard

Passionate Game Developer and Engineer Bringing
Virtual Worlds to Life through Advanced Programming

Rue de la Jonction 4
1400 Yverdon-les-Bains
078 647 00 28
quentin.guignard@outlook.com

Experiences

OZWE Games — *Technical Artist / Developer*

Internship, June 2023 - August 2023

Particle Effects development (PopcornFx) for Meta Horizon World.
Gameplay programming and game performance optimization.

EPFL, Scala Center — *Teaching Assistant*

February 2020 - June 2023

Q&A forum management

EPFL, LARA Laboratory — *Teaching Assistant*

Spring semester 2021

CS-206 Parallel and concurrency

EPFL, Digital Humanities Laboratory — *Teaching Assistant*

Spring semester 2020

CS-213 Informatique visuelle

City Cleaning Office, Orbe VD — *Work Assistant*

August 2019 - summer job

Education

ETH — *Master in Computer Science*

2021 - 2024

Major in Visual computing, Minor in Systems software

EPFL — *Bachelor in Computer Science*

2017 - 2021

Gymnase de Cheseaux-Noréaz — *Maturity in Biology and Chemistry*

2014 - 2017

Skills

Proficient in low level software
(Rust/C++)

Experienced with Unity (C#),
Bevy Game Engine (Rust) and
Particle Effects (PopcornFX)

AI development (autonomous
agents, LLM, prompt
engineering, machine learning)

Strong knowledge in Distributed
Systems and Concurrent
Programming

Proficient in Python (data
analysis, plotting, computer
vision) and tools automation

Mathematical knowledge
(Geometry, Linear algebra and
Optimization)

Hobbies

Personal coding projects
[Youtube channel](#), [blog](#) and
[github](#).

Video games (currently casual
gaming)

Acrylic Painting

Sport: running (maratonian)
and calisthenics

Languages

French - mother tongue
English - proficient
German - B2
Russian - A1.1

Projects Portfolio

[Andrometica Cynus](#) (2024) — Master Thesis, Game

Research Game prototype exploring LLM as game input method. Written in Rust with the Bevy Engine.

[BackRooms Level 2](#) (2023) — Game

Unity horror game based on the “backroom” universe. Features an AI character chasing the player.

[Raytracer](#) (2022) — Software Development

Individual ETH Student project. Features CPU raytracer (C++), light physics, geometry and scene modelization.

[LevelkGL](#) (2020-2022) — Software Development

C++ OpenGL wrapper used to learn and discover the basics of graphics programming.

[Wave Function Collapse](#) (2022) — Software Development

Implementation of the wave function collapse algorithm in C++ with rendering (OpenGL).

[Real-Time Fluid Simulation](#) (2022) — Software Development

C++ constraint solver, water rendering (OpenGL & GLSL) based on Screen Space Fluid. [Source](#).

[Caterine](#) (2022) — Game

Student group project. I participated in the interop C#/C++, sand simulation (C++), characters animations, voxel art (Magika Voxel) and Monogame (C#) development.

[SuperVoxel](#) (2021) — Bachelor Thesis

Exploring techniques to physically simulate voxel objects fractures in a video game context.

[Paradisland](#) (2020) — Digital Experience

EPFL student project featuring water shader, outlining effect shader, shadows (GLSL) rendering and scene modelization using Blender.